

TrustCXR: An Evidence-Governed and Uncertainty-Aware Chest X-ray Research System

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Abstract. Chest X-ray models can produce useful population-level signals while remaining vulnerable to dataset shift, ambiguous labels, and unsupported clinical interpretation. TrustCXR is a frozen, local research system designed around explicit evidence boundaries rather than autonomous diagnosis. The pipeline combines view and technical-quality assessment, a fourteen-label chest-radiograph classifier, model-specific predictive-uncertainty evidence, conservative evidence integration, deterministic research reporting, structured verification, and a defer-first decision policy for expert review. The system preserves provenance and unavailable evidence instead of converting missing support into a positive claim or a contradiction. Frozen internal evaluation of the accepted classifier yielded macro AUROC 0.7287, macro AUPRC 0.1540, and F1 at 0.5 of 0.2072 on 17,061 records from 4,715 patients. These values are research evidence from a governed development/test program, not clinical probabilities or validation of deployment. The release includes bounded local PNG/JPEG inference, a FastAPI service, and a static research interface; it does not include clinical diagnosis, reliable positive lesion localization, external validation, active learning, or language-model reporting.

Keywords: chest X-ray; medical imaging; multi-label classification; predictive uncertainty; evidence verification; reproducible AI; expert review

Research use only. TrustCXR is not a medical diagnostic system and requires expert review.

1. Introduction

Chest radiography is widely used for thoracic assessment, and deep learning has made multi-label image classification a practical research direction. Large datasets such as ChestX-ray8/14 and CheXpert enabled learning from weak or uncertain report-derived labels [1, 2]. However, a classifier score alone does not establish a diagnosis, a calibrated clinical probability, or a reliable explanation of where a finding is located. Dataset shift, label semantics, acquisition variation, and incomplete spatial supervision can all make an apparently confident output unsafe to overinterpret.

TrustCXR addresses this systems problem as a research engineering exercise. Its objective is not to automate radiology, but to place a frozen vision pipeline inside an explicit evidence contract: each component has an accepted scope, missing support remains visible, and a deterministic verifier and decision layer can defer a draft when evidence is insufficient. This design is motivated by the broader observation that useful predictive uncertainty is difficult to maintain under distribution shift [3].

The contributions of the frozen core are: (1) a patient-governed multi-stage chest X-ray research pipeline; (2) frozen view/quality and fourteen-label classification with model-specific uncertainty evidence; (3) conservative evidence packaging, deterministic report verification, and defer-first research decision support; and (4) a reproducible local application whose unsupported capabilities are explicitly withheld.

2. Related Work

Weakly supervised thoracic classification was established at scale by ChestX-ray8, which paired radiographs with report-derived labels and also studied weak localization [1]. CheXpert extended this setting with uncertainty labels and expert comparison for selected findings [2]. TrustCXR uses these datasets as governed research sources, while preserving their different label semantics rather than treating them as interchangeable ground truth.

The accepted classifier family uses densely connected convolutional representations, a design that promotes feature reuse and short information paths [4]. View/quality and multiview experiments also draw on the practical need to control acquisition context before interpreting a radiograph. For interpretability, Grad-CAM is relevant background for a future attribution study [5]; it is not implemented in the frozen core, and an attribution map would not by itself establish lesion localization.

3. Data Governance and Study Design

Raw datasets remain local and intentionally untracked. Development splits are patient-safe where governed, and locked-test protection is preserved. Dataset roles, label limitations, and external-validation eligibility are summarized in Table 1. The accepted classifier evidence uses a frozen internal cohort; no independent external cohort satisfied the release criteria, so external validation was not performed.

4. TrustCXR Framework

A. View and Technical-Quality Assessment

The perception layer identifies AP, PA, or LATERAL view and produces a non-clinical technical-quality proxy. The proxy is a research control on image usability, not a statement about diagnostic adequacy. On the CheXpert Small internal test, the frozen view head achieved macro F1 0.9861 and the proxy achieved accuracy 0.9994; these values are task-specific and are not compared directly with classifier metrics.

B. Multi-Label Classification

The frozen classifier returns one research model score for each of the following labels: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia. Scores are responses of the model, not calibrated clinical disease probabilities and not diagnoses. The classifier was selected without post-test tuning and retained under a fixed preprocessing and label-order contract.

Table 1: Governed dataset roles and final status. Raw images and identifiers are not redistributed.

Dataset	Role	Governance and label scope	Final status
NIH ChestX-ray14	Classification; shared development evidence	Patient-safe split evidence; fourteen dataset-specific labels	Development use; not external
CheXmask	Pseudo anatomy evidence	Quality-filtered masks; not lesion ground truth	Limited anatomy research
RSNA Pneumonia	Localization baseline	Lung-opacity boxes; narrower than fourteen-label scope	Research baseline only
CheXpert Small	View/quality and multiview studies	Different label contract; governed paired splits	Internal experiments; not external
Audited candidates	External-validation review	Identity, licensing, or semantics unresolved	No eligible external cohort

C. Predictive Uncertainty

TrustCXR reports predictive uncertainty only where the frozen calibration contract applies. It does not claim epistemic uncertainty, validated out-of-distribution detection, or clinical certainty. This distinction is important because uncertainty quality can degrade under dataset shift even when in-distribution performance appears stable [3].

D. Evidence Integration

Structured outputs are packaged with provenance and scope. Pseudo-anatomy and localization evidence can support limited research statements, but missing spatial evidence is not treated as contradiction. The fusion layer therefore has a maximum partially-supported interpretation and cannot provide reliable positive lesion localization or finding laterality.

E. Deterministic Reporting and Verification

The report renderer converts structured evidence into a template-bound research draft. A verifier checks claims against accepted structured and provenance evidence, preserving unavailable states and restricting contradiction to explicit governed conflicts. A conservative decision layer applies the precedence DEFER over revise over acceptance for expert review. Acceptance means that a research draft is ready for expert review; it is never clinical approval.

5. Experimental Protocol

The final classifier result was taken from the frozen internal evaluation of 17,061 records from 4,715 patients. Selection was completed on validation evidence, and locked-test records were not re-opened for this paper. Metrics from the separate frontal-only study and the localization baseline are reported with their own cohort qualifications rather than combined into a single ranking. The paper reports existing values only; no metric was recomputed for publication.

6. Results and Analysis

Table 2 presents the principal frozen classifier result with sensible display precision. The model provides useful research discrimination evidence, but the modest AUPRC and absence of external validation limit any interpretation beyond the governed cohort.

Table 2: Frozen fourteen-label classifier result. Values are rounded for presentation; full-precision values remain in release evidence.

Metric	Value
Macro AUROC	0.7287
Macro AUPRC	0.1540
F1 @ 0.5	0.2072
Test records	17,061
Test patients	4,715

7. System Implementation and Reproducibility

The release provides a local FastAPI service and a static research UI for bounded PNG/JPEG review. The accepted runtime executes the GPU models sequentially, then performs deterministic CPU-side evidence, report, verification, and decision processing. The interface presents current-image model scores and a concise deterministic explanation without an LLM, external requests, browser persistence, or patient workflow.

The reproducibility contract is Windows 11 with Python 3.12.10, PowerShell orchestration, PyTorch cu130, and a governed NVIDIA GPU environment. Seven checkpoint/configuration pairs are retained with SHA-256 evidence. Controlled seeds and frozen configurations support reproducible research execution, but bit-identical CUDA training is not guaranteed. The core release is identified internally as the frozen research release and the UI as the frozen core review interface; these identifiers are implementation traceability, not scientific claims.

8. Limitations, Ethics, and External Validation

The system is research use only, is not a medical diagnostic system, and requires expert review. It does not provide treatment recommendations, autonomous clinical release, severity, temporal comparison, or validated OOD detection. The localization baseline has no accepted operating threshold and does not support reliable positive lesion localization; absence of localization must not contradict classifier evidence. Selective prediction was not accepted, and the deterministic triage layer is defer-first.

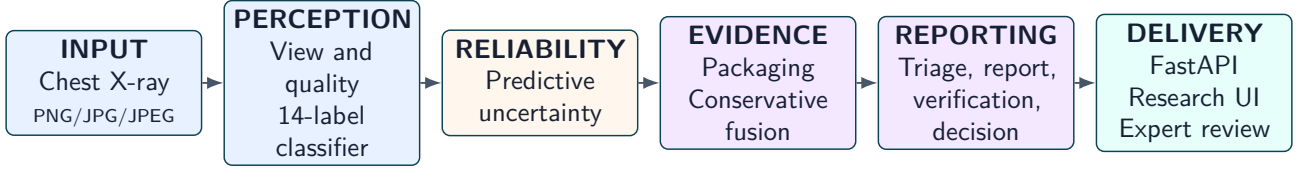


Figure 1: Overview of the frozen TrustCXR research pipeline. Stage identifiers are retained in the release evidence for traceability but are not user-facing capability claims.

Table 3: Additional frozen evidence. Cohorts and contracts differ; rows are not superiority comparisons.

Component	Cohort	Metric	Result	Scientific qualification
View assessment	CheXpert Small internal test	Macro F1	0.9861	AP/PA/LATERAL view task
Technical-quality proxy	CheXpert Small internal test	Accuracy	0.9994	Non-clinical proxy target
Pseudo anatomy	NIH CheXmask internal test	Dice / IoU	0.9714 / 0.9444	Pseudo anatomy, not lesion ground truth
Localization baseline	RSNA validation	Small-lesion sensitivity	0.0361	No acceptable operating point
Frontal-only classifier	CheXpert exact-pair test	AUROC / AUPRC	0.8467 / 0.8593	Different labels/cohort; not comparable to Stage 9

Table 4: Capability and limitation matrix for the frozen core.

Capability	Status	Interpretation
Fourteen-label classification	Supported	Frozen research model signals; not diagnoses
Predictive uncertainty	Limited	Model-specific predictive evidence only
Lesion localization	Withheld	No reliable positive localization or laterality
OOD detection	Withheld	Not validated
Temporal comparison / severity	Withheld	Not supported
DICOM	Limited	Synthetic/non-patient interoperability only
External validation	Not performed	No generalizability claim

DICOM interoperability is limited to synthetic/non-patient, single-frame grayscale fixtures under the governed transfer-syntax and photometric contracts. Real-patient DICOM handling and a browser DICOM viewer are outside scope. Human feedback and active learning were withheld because no governed expert-feedback data existed. External validation was not performed because no independent cohort satisfied identity, licensing, label, and cohort-construction requirements. Consequently, the release makes no clinical generalizability, prospective, multi-institutional, or deployment claim.

For release traceability, the frozen evidence disposition is recorded as `EXTERNAL_VALIDATION_NOT_PERFORMED`; this is an evidence status, not a claim of external evaluation.

Privacy requires that patient identifiers, PHI, raw private DICOM metadata, credentials, and raw datasets remain outside the repository. The project is a reproducible research pipeline, not production clinical MLOps.

9. Future Research Directions

Post-release work is documented separately and is not implemented in this core paper. A first track may study class-specific visual attribution (for example, Grad-CAM) while describing maps as model attention rather than validated lesion location [5]. A second track would require independently governed spatial annotations for true pathology localization. A third could evaluate a grounded language model that consumes only structured evidence, keeps the deterministic renderer as a baseline, and remains behind verifier and expert-review gates. A multimodal VLM study would require a separate explicit authorization and comparison against the frozen vision pipeline.

10. Conclusion

TrustCXR demonstrates a conservative way to package chest X-ray research models for expert-oriented review. Its contribution is the combination of frozen perception and classification, predictive-uncertainty evidence, provenance-aware evidence handling, deterministic reporting and verification, and a defer-first decision policy within a reproducible local application. The evidence supports a research system with explicit boundaries, not a clinical diagnostic product. External validation, reliable positive localization, active learning, language-model reporting, and other extensions remain future work and are not required for the validity of the frozen core release.

References

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Appendix A. Reproducibility Artifacts

The frozen manuscript is accompanied by an evidence trail rather than a claim that the paper replaces the underlying records. The final claims matrix, metrics table, dataset-use summary, technical report, reproducibility lock, and post-release roadmap remain the authoritative project artifacts. They preserve exact hashes, cohort qualifications, withheld dispositions, and reconstruction instructions without redistributing raw data or patient identifiers.

Artifact	Role
Claims matrix	Supported, limited, withheld, and prohibited-claim contract
Metrics table	Frozen values with cohort and qualification context
Dataset-use summary	Governance, split, label, and external-eligibility record
Environment lock	Windows/CUDA reconstruction contract
Architecture asset	Compact vector description of the frozen core
Extension roadmap	Optional, post-release research only; not implemented

Reconstruction is anchored by the final Windows/CUDA environment lock and its installation protocol. The local service accepts bounded raster input, validates decoding, runs the view/quality and classifier models sequentially on the governed GPU, and releases model residency before deterministic downstream processing. This ordering is an operational reproducibility constraint, not a claim of production deployment.

The evidence boundary is equally important: raw datasets remain outside Git, patient-safe split records and locked-test protections are preserved, and the release does not include external validation, human-feedback acquisition, attribution maps, pathology localization, or language-model reporting. These boundaries are part of the reproducibility result because they define what another authorized researcher is, and is not, being asked to reconstruct.